

Milky Way foreground-screen stratification in a SPARC/HECATE residual audit

Abstract

The Milky Way foreground is an observational layer, not merely a nuisance. Zone of Avoidance work with JWST/NIRCam has recently shown that faint background galaxies can be recovered through heavily contaminated Galactic regions, while also emphasizing that extinction, stellar crowding, and confusion noise shape extragalactic visibility. Motivated by this, we define a deliberately simple foreground-screen diagnostic for the public SPARC residual-disturbance audit: Galactic latitude computed from HECATE RA/DEC for exact-name SPARC overlaps. Using the provisional screen $|b| \leq 24^\circ$, the packet selects 18 systems. In the reviewed A/C subset, this screen is C-dominant: A=1 and C=6, with 11 ambiguous B systems. The A/C enrichment check is not statistically decisive ($p = 0.295343139746$), so the result is best read as an observer-screen audit and preregistration target, not a detection.

Motivation

Nilo-Castellon et al. identify 102 galaxies in a JWST/NIRCam field in the Zone of Avoidance and describe Galactic extinction, stellar crowding, and confusion noise as historical limits on background-galaxy detection. This motivates a conservative question for SPARC residual audits: do systems projected nearer to the Milky Way plane occupy a visibly different label/observability stratum?

The point of this note is not to explain galaxy dynamics. It is to preserve a small, reproducible foreground-screen packet that can be tested later with better extinction maps, star-count fields, and independent galaxy samples.

Data And Construction

Inputs:

- HECATE v1.1 RA/DEC for exact-name SPARC overlaps.
- Residual-blind A/B/C labels from the public Paper 1 SPARC residual-disturbance packet.

- Optional residual summary fields from the public disturbance-inference packet.

The script converts HECATE equatorial coordinates to Galactic longitude and latitude using the standard J2000 north Galactic pole constants. The primary screen is:

low_latitude_screen = |b| <= 24 deg

This threshold is not claimed as fundamental. It is a transparent reconstruction of the remembered 18-object screen and should be stress-tested in future work.

Main Diagnostic

Metric	Value	Interpretation
hecate_sparc_overlap_rows	120	exact-name SPARC/HECATE overlaps with RA/DEC
low_latitude_threshold_deg	24.0	preliminary foreground-screen proxy
low_latitude_total	18	b <= 24 deg
low_latitude_A	1	reviewed calm/regular class count
low_latitude_B	11	ambiguous class count
low_latitude_C	6	reviewed disturbed class count
high_latitude_A	15	b > 24 deg
high_latitude_B	57	b > 24 deg
high_latitude_C	30	b > 24 deg
ac_one_sided_c_enrichment_p	0.295343139746	diagnostic only; not a detection

Threshold Scan

AbsGalacticLatitudeThresholdDeg	N_total	N_AC	N_A	N_B	N_C
5	0	0	0	0	0
7.5	0	0	0	0	0
10	1	0	0	1	0
12	2	0	0	2	0
15	8	2	0	6	2
20	13	5	0	8	5
22	15	6	1	9	5
23	17	6	1	11	5
24	18	7	1	11	6
24.5	18	7	1	11	6
25	19	8	1	11	7
26	20	9	1	11	8
27	21	10	2	11	8
28	21	10	2	11	8
29	23	12	2	11	10
30	25	13	3	12	10

The 18 Low-Latitude Systems

GalaxyName	Class	AbsGalacticLatitudeDeg	GalacticLongitudeDeg	HecateDistanceMpc	Projection_RMS	ResidualDisturbanceScore_v01
UGC03205	B	8.206377	342.903037	54.37		
NGC6946	B	11.672188	60.144887	5.516		
UGC11557	B	12.814719	60.883601	23.66		
UGC00731	C	13.149452	29.723502	13.03		
NGC6674	B	13.919223	101.271460	54.11		

UGC02885	B	14.050142	356.283096	71.12			
UGC02916	C	14.194665	19.057928	66.11			
ESO563-G021	B	14.397417	270.996728	79.44			
NGC0891	C	17.415249	15.479606	9.111			
NGC1003	B	17.545110	11.859346	10.76			
NGC2915	C	18.357173	223.897427	4.03			
UGC12632	C	19.311054	49.092790	9.204			
UGC02259	B	19.796208	8.718768	9.954			
NGC7331	A	20.724275	62.141728	14.4			
NGC6789	B	21.517771	60.889695	3.267			
NGC0801	B	22.456867	17.644924	55.53			
UGC11455	B	22.844922	52.115383	85.75			
NGC3109	C	23.070239	253.762265	1.291			

Interpretation

The result is directionally interesting because the low-latitude screen is C-heavy among systems that already have reviewed A/C labels. However, the sample is small and B-dominated. A one-sided A/C hypergeometric enrichment diagnostic returns $p = 0.295343139746$, which is not a discovery-level result.

Residual-Signal Stress Test

The natural follow-up question is whether the 18-object screen is also a good place to look for a residual projection signal. We therefore join the foreground screen to the public Paper 1 residual summary and test two distinct possibilities:

1. A residual-scatter excess.
2. A signed residual offset.

The current packet does **not** support a simple RMS-excess claim. The median ``rms_log_tpg`` in the low-latitude screen is slightly lower than the high-latitude comparison set. The more interesting candidate is instead a signed offset: ``mean_log_residual_tpg`` and ``outer_mean_log_residual_tpg`` shift positive in the low-latitude screen. This is exactly the sort of weak observer-screen candidate that should be preregistered before any stronger interpretation.

Metric	Value	Interpretation
---	---	---
residual_summary_available	true	public Paper 1 residual summary joined to foreground screen
residual_join_rows	120	HECATE/SPARC rows with residual summary
residual_low_screen_rows	18	b <= 24 deg
residual_high_screen_rows	102	b > 24 deg
rms_log_tpg_low_median	0.140050995433	low-latitude foreground screen
rms_log_tpg_high_median	0.146757312311	higher-latitude comparison set
rms_log_tpg_low_minus_high_median	-0.00670631687742	diagnostic difference; not a detection
weighted_rms_log_tpg_low_median	0.116555387387	low-latitude foreground screen

```

| weighted_rms_log_tpg_high_median | 0.137194953464 | higher-latitude comparison set |
| weighted_rms_log_tpg_low_minus_high_median | -0.0206395660763 | diagnostic
difference; not a detection |
| mean_log_residual_tpg_low_median | 0.0477160372602 | low-latitude foreground screen |
| mean_log_residual_tpg_high_median | -0.039649763687 | higher-latitude comparison set
|
| mean_log_residual_tpg_low_minus_high_median | 0.0873658009472 | diagnostic
difference; not a detection |
| outer_mean_log_residual_tpg_low_median | 0.00914315295169 | low-latitude foreground
screen |
| outer_mean_log_residual_tpg_high_median | -0.0241318063717 | higher-latitude
comparison set |
| outer_mean_log_residual_tpg_low_minus_high_median | 0.0332749593234 | diagnostic
difference; not a detection |
| mean_err_vobs_kms_low_median | 5.77818965517 | low-latitude foreground screen |
| mean_err_vobs_kms_high_median | 5.1711037234 | higher-latitude comparison set |
| mean_err_vobs_kms_low_minus_high_median | 0.607085931768 | diagnostic difference; not
a detection |
| max_radius_kpc_low_median | 25.32 | low-latitude foreground screen |
| max_radius_kpc_high_median | 12.935 | higher-latitude comparison set |
| max_radius_kpc_low_minus_high_median | 12.385 | diagnostic difference; not a
detection |
| n_points_low_median | 22 | low-latitude foreground screen |
| n_points_high_median | 13 | higher-latitude comparison set |
| n_points_low_minus_high_median | 9 | diagnostic difference; not a detection |
| interpretation | candidate_signed_projection_offset_not_rms_excess_detection | RMS
median is not elevated in the 18-object screen; signed residual medians shift positive |

```

Matched-Control Preview

As a first guardrail, each low-latitude galaxy is greedily matched to a unique higher-latitude control by log HECATE distance, log radial extent, and log point count. This is only a preview; it is not a covariance-aware likelihood or an extinction-aware analysis.

```

| Metric | Value | Interpretation |
| --- | --- | --- |
| matched_pair_count | 18 | greedy unique high-latitude controls matched on log
distance, max radius, and point count |
| matched_rms_log_tpg_median_low_minus_high | -0.00641394691747 | paired diagnostic
difference; not a detection |
| matched_rms_log_tpg_positive_pairs | 9/18 | sign count across matched pairs |
| matched_weighted_rms_log_tpg_median_low_minus_high | 0.0158783211739 | paired
diagnostic difference; not a detection |
| matched_weighted_rms_log_tpg_positive_pairs | 10/18 | sign count across matched pairs
|
| matched_mean_log_residual_tpg_median_low_minus_high | 0.0100950616694 | paired
diagnostic difference; not a detection |
| matched_mean_log_residual_tpg_positive_pairs | 9/18 | sign count across matched pairs
|
| matched_outer_mean_log_residual_tpg_median_low_minus_high | 0.0105481845933 | paired
diagnostic difference; not a detection |
| matched_outer_mean_log_residual_tpg_positive_pairs | 9/18 | sign count across matched
pairs |

```

The correct reading is therefore:

The Milky Way foreground screen is a real observability stratum. In this SPARC/HECATE packet, a reconstructed 18-object low- $|b|$ screen is C-dominant among reviewed A/C labels, but the result remains a preregistered audit target rather

than a physical inference.

The residual follow-up does not show a robust RMS-excess detection. It does show a candidate signed residual-offset direction that is worth freezing as the next observer-screen test.

Next Tests

1. Replace the latitude-only proxy with $E(B-V)$, A_V , star-count density, and local confusion metrics.
2. Freeze the screen thresholds before residual inspection.
3. Re-run the audit on independent low-latitude and high-latitude galaxy samples.
4. Separate foreground observability effects from intrinsic disturbance evidence.
5. Report all A/B/C transitions under threshold scans rather than selecting only the most favorable boundary.

Claim Boundary

This note does not claim new dynamics, a physical detection, or proof of any private parent theory. It is a public observer-screen method note.

References

- J. L. Nilo-Castellon et al., *Faint galaxies in the Zone of Avoidance revealed by JWST/NIRCam*, arXiv:2510.12488, A&A; 704, A209 (2025).
- SPARC residual-disturbance Paper 1 public reproducibility package.
- HECATE v1.1 public catalog.